

NETWORKWALKS

WEEK 3 — PASSWORD SECURITY ASSESSMENT

Public Portfolio Edition

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This public edition summarizes the technical assessment while intentionally excluding recovered laboratory credentials, crackable password hashes, original locked PDFs, and CTF completion secrets.

1. Executive Summary

The Week 3 assessment tested offline PDF password recovery using John the Ripper with RockYou and browser-based Networkwalks password-cracking tools. JtR recovered all three supplied lab artifacts. Networkwalks recovered PDF2 and PDF3 with its captured built-in dictionary; PDF1 exhausted the bounded dictionary without a match.

2. Key Technical Observation

PDF1 demonstrates candidate-set coverage. A bounded dictionary can fail to recover a password that a larger dictionary subsequently recovers. Therefore, a dictionary attack returning “No match” is not evidence that a password is strong or uncrackable.

3. PM1 — John the Ripper

The local workflow used pdf2john.pl to extract PDF password-verification material, JtR to perform the dictionary attack, john --show for recovery confirmation, and qpdf/Poppler tooling for independent PDF1 validation and content inspection.

4. PM2 — Networkwalks Tools

The browser workflow used the Networkwalks Hash Calculator to extract a crackable PDF representation and the Networkwalks Password Cracker to test dictionary candidates. The workflow reduces setup friction but the captured built-in candidate set is bounded.

5. Defensive Interpretation

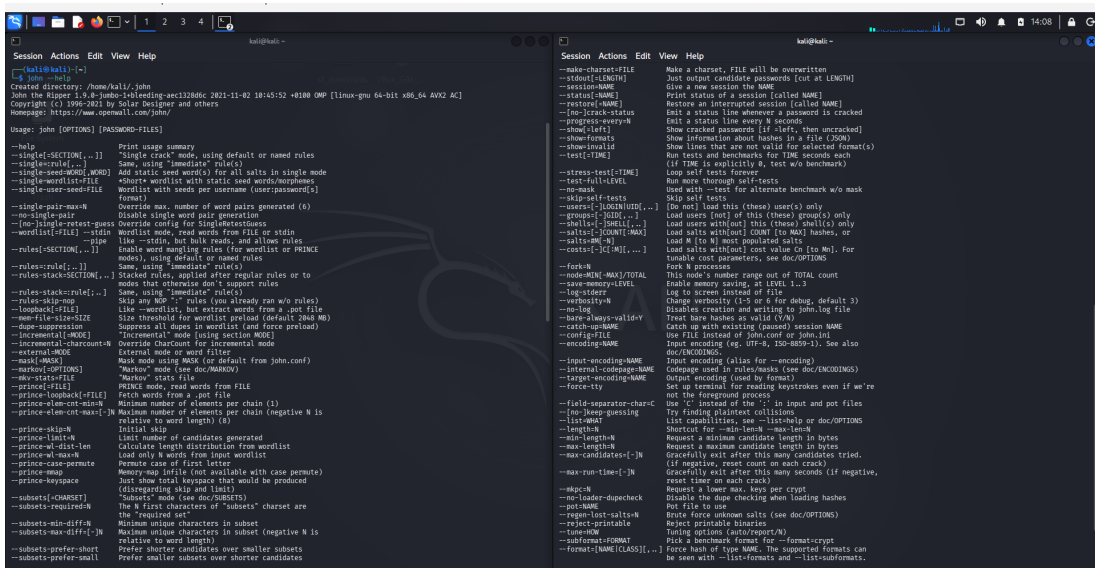
Offline verifier exposure changes the threat model because password candidates can be tested without online lockout. Defenses should include strong unique passwords, breached-password screening, adaptive password hashing, MFA, credential-reuse prevention, and protection of sensitive credential-verification material.

6. Evidence Handling

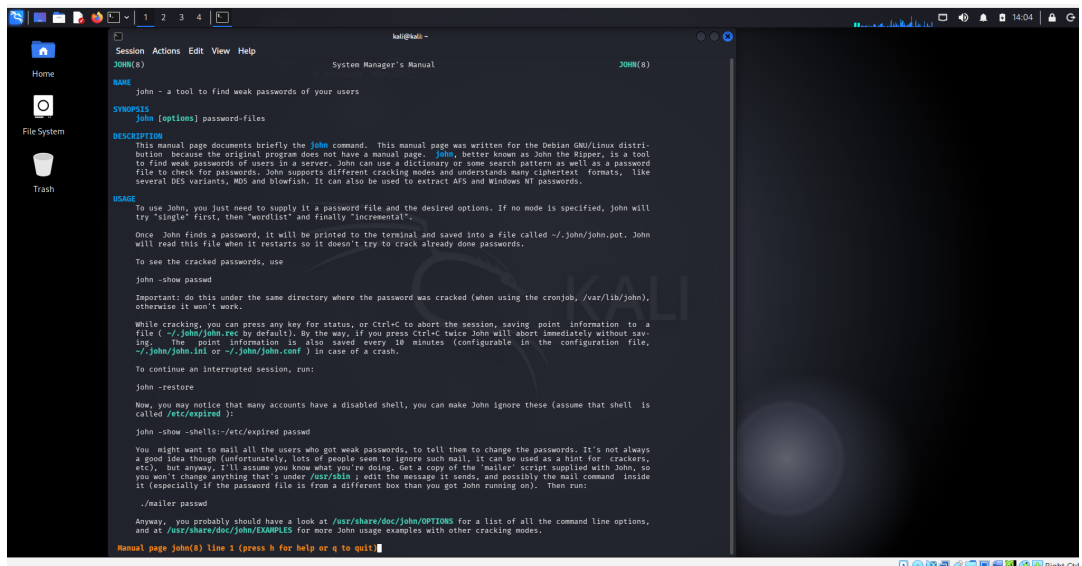
Public portfolio material omits passwords and crackable hashes. Full private evidence remains separate from this repository-ready package.

Artifact	JtR + RockYou	Networkwalks built-in	Observed significance
PDF1	Recovered	Not recovered	Candidate coverage changed result
PDF2	Recovered	Recovered at 91/100	Within bounded list
PDF3	Recovered	Recovered at 35/100	Within bounded list

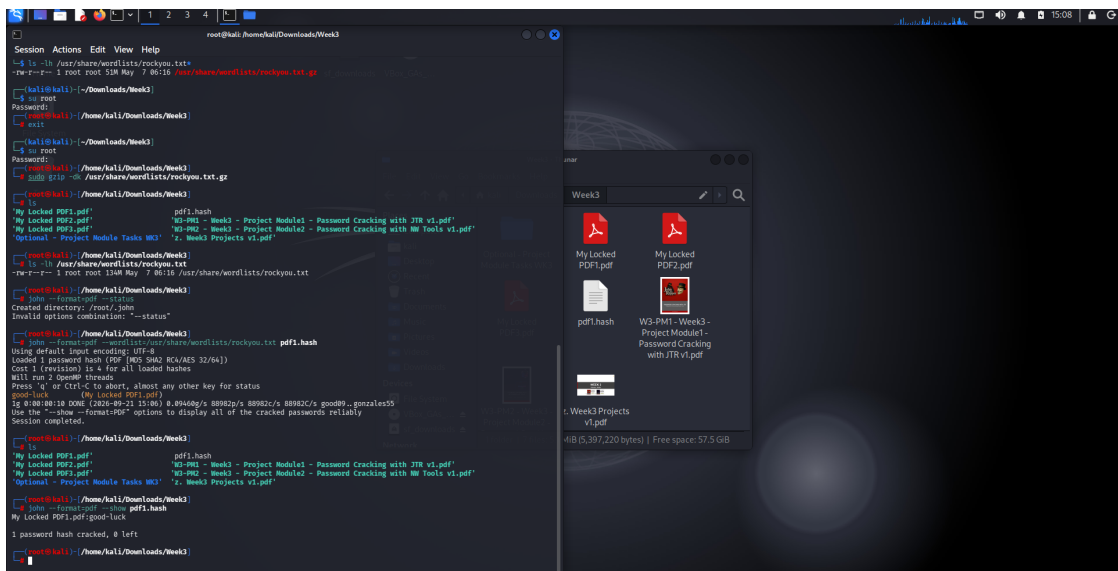
Selected Public Evidence



JtR capability context



Dictionary methodology



RockYou preparation

```
root@kali: /home/kali/Downloads/Week3
Session Actions Edit View Help
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 1331
Download size: 1,262 KB
Space needed: 4,018 KB / 61.6 GB available

Continued from y
Get:1 http://kali.download/kali kali-rolling/main amd64 libqpdf3 amd64 12.4.1-1 [1,100 KB]
Get:2 http://kali.download/kali kali-rolling/main amd64 qpdf amd64 12.4.1-1 [163 KB]
Fetched 1,262 KB in 3s (158 KB/s)
Selecting previously unselected package libqpdf3:amd64.
(Reading database ... 43277 files and directories currently installed.)
Preparing to unpack .../libqpdf3_12.4.1-1_amd64.deb...
Unpacking libqpdf3:amd64 (12.4.1-1)...
Selecting previously unselected package qpdf.
Preparing to unpack .../qpdf_12.4.1-1_amd64.deb...
Unpacking qpdf (12.4.1-1)...
Setting up libqpdf3:amd64 (12.4.1-1)...
Setting up qpdf (12.4.1-1)...
Processing triggers for libc-bin (2.40-10)...
Processing triggers for man-db (2.13.1-1)...
Processing triggers for kali-menu (2026.2.6)...

root@kali: /home/kali/Downloads/Week3
# qpdf
root@kali: /home/kali/Downloads/Week3
qpdf --password="good-luck" --verbose --show-encryption "My Locked PDF1.pdf"
qpdf: unrecognized argument --show-encryption My Locked PDF1.pdf

For help:
qpdf --help-usage      usage information
qpdf --help-topic      help on a topic
qpdf --help-option     help on an option
qpdf --help            general help and a topic list

root@kali: /home/kali/Downloads/Week3
# qpdf --password="good-luck" --verbose --show-encryption "My Locked PDF1.pdf"
q & 4
P & 4
User password = good-luck
Supplied password is owner password
Supplied password is user password
extract for accessibility: allowed
extract for any purpose: allowed
print low resolution: allowed
print high resolution: allowed
modify forms: allowed
modify document assembly: not allowed
modify fonts: allowed
modify annotations: allowed
modify other: allowed
modify anything: not allowed
stream encryption method: AESv2
string encryption method: AESv2
file encryption method: AESv2

root@kali: /home/kali/Downloads/Week3
Session Actions Edit View Help
PDF(1)
User Commands
qpdf(1)
NAME
qpdf - PDF transformation software
SYNOPSIS
qpdf [ options ] infilename [ outfilename ]
DESCRIPTION
The qpdf program is used to convert one PDF file to another equivalent PDF file. It is capable of performing a variety of transformations such as linearization (also known as web optimization or fast web viewing), encryption, and decryption of PDF files. It also has many options for inspecting or checking PDF files, some of which are useful primarily to PDF developers.
For a summary of qpdf's options, please run qpdf --help. A complete manual can be found at https://qpdf.readthedocs.io.
In some distributions, you can find offline versions of the qpdf manual in /usr/share/doc/qpdf. This may require installing an additional package, such as qpdf-doc in Debian-based Linux distributions. You can also download offline documentation from https://qpdf.readthedocs.io. For more information, see README-doc.txt in the source distribution.
USAGE (basic invocation)
Read a PDF file, apply transformations or modifications, and write a new PDF file.
Usage: qpdf [infile] [options] [outfile]
OR qpdf --help[=topic|--option])
- infile, options, and outfile may be in any order as long as infile precedes outfile.
- Use --empty in place of an input file for a zero-page, empty input
- Use --replace-input in place of an output file to overwrite the input file with the output
- outfile may be - to write to stdout; reading from stdin is not supported
- filename is an argument file; each line is treated as a separate command-line argument
- @- may be used to read arguments from stdin
- Later options may override earlier options if contradictory
Related Options:
--empty -- use empty file as input
Use in place of infile for an empty input. Especially useful with --pages.
--replace-input -- overwrite input with output
Use in place of outfile to overwrite the input file with the output.
--job-json-file -- job JSON file
--job-json-file=file
Manual page qpdf(1) line 1 (press h for help or q to quit)
```

qpdf tooling

```
(root@kali)-[/home/kali/Downloads/Week3]
# pdftotext "My Locked PDF1_decrypted.pdf" - | head -40

(root@kali)-[/home/kali/Downloads/Week3]
# pdftimages -list "My Locked PDF1_decrypted.pdf"
page num type width height color comp bpc enc interp object ID x-ppi y-ppi size ratio
1 0 image 1055 1491 rgb 3 8 jpeg no 7 0 128 128 265K 5.7%

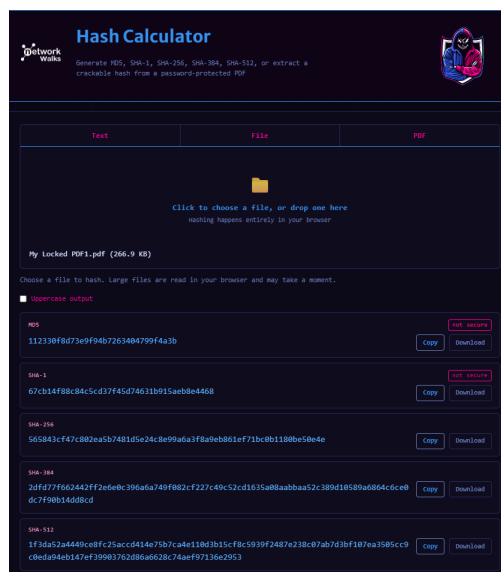
(root@kali)-[/home/kali/Downloads/Week3]
# pdftoppm -png -f 1 -singlefile \
"My Locked PDF1_decrypted.pdf" \
"My_Locked_PDF1_page1"

(root@kali)-[/home/kali/Downloads/Week3]
# ls -lh My_Locked_PDF1_page1.png
-rw-rw-r-- 1 root root 1.5M Sep 21 15:52 My_Locked_PDF1_page1.png

(root@kali)-[/home/kali/Downloads/Week3]
# xdg-open My_Locked_PDF1_page1.png

(root@kali)-[/home/kali/Downloads/Week3]
```

PDF inspection



Networkwalks workflow

PM2-BPDF1-Bounded Dictionary Result
Networkwalks Password Cracker
Built-in dictionary exhausted without a match
Observed run: 100 / 100 candidates@approximately 8 pws
Interpretation: bounded negative result; larger candidate coverage may change the outcome
Public portfolio evidence @alternative credentials and crackable hashes intentionally omitted

PDF1 bounded dictionary result

PM2-BPDF2-BDictionary Recovery
Networkwalks Password Cracker
Password recovered during dictionary attack
Observed run: 91 / 100 candidates@approximately 9 pws
Credential value intentionally omitted from public evidence
Public portfolio evidence @alternative credentials and crackable hashes intentionally omitted

PDF2 recovery result

PM2-BPDF3-BDictionary Recovery
Networkwalks Password Cracker
Password recovered during dictionary attack
Observed run: 38 / 100 candidates@approximately 4 pws
Credential value intentionally omitted from public evidence
Public portfolio evidence @alternative credentials and crackable hashes intentionally omitted

PDF3 recovery result